

Question 1 | Correct | Mark 1.00 out of 1.00

Given two arrays `array_One[]` and `array_Two[]` of same size `N`. We need to first rearrange the arrays such that the sum of the product of pairs (1 element from each) is minimum. That is $\text{SUM}(A[i] * B[i])$ for all `i` is minimum.

For example:

Input	Result
3	28
1	
2	
3	
4	
5	
6	

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int compare_asc(const void *a, const void *b) {
5     long long diff = (long long)*(int *)a - (long long)*(int *)b;
6     if (diff < 0) return -1;
7     if (diff > 0) return 1;
8     return 0;
9 }
10
11 int compare_desc(const void *a, const void *b) {
12     long long diff = (long long)*(int *)b - (long long)*(int *)a;
13     if (diff < 0) return -1;
14     if (diff > 0) return 1;
15     return 0;
16 }
17
18 int main() {
19     int n;
20     if (scanf("%d", &n) != 1 || n <= 0) return 0;
21
22     int A[n], B[n];
23
24     for (int i = 0; i < n; i++) {
25         scanf("%d", &A[i]);
26     }
27
28     for (int i = 0; i < n; i++) {
29         scanf("%d", &B[i]);
30     }
31
32     qsort(A, n, sizeof(int), compare_asc);
33     qsort(B, n, sizeof(int), compare_desc);
34
35     long long min_sum = 0;
36     for (int i = 0; i < n; i++) {
37         min_sum += (long long)A[i] * B[i];
38     }
39
40     printf("%lld\n", min_sum);
41     return 0;
42 }
```

	Input	Expected	Got	
✓	3 1 2 3 4 5 6	28	28	✓
✓	4 7 5 1 2 1 3 4 1	22	22	✓
✓	5 20 10 30 10 40 8 9 4 3 10	590	590	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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